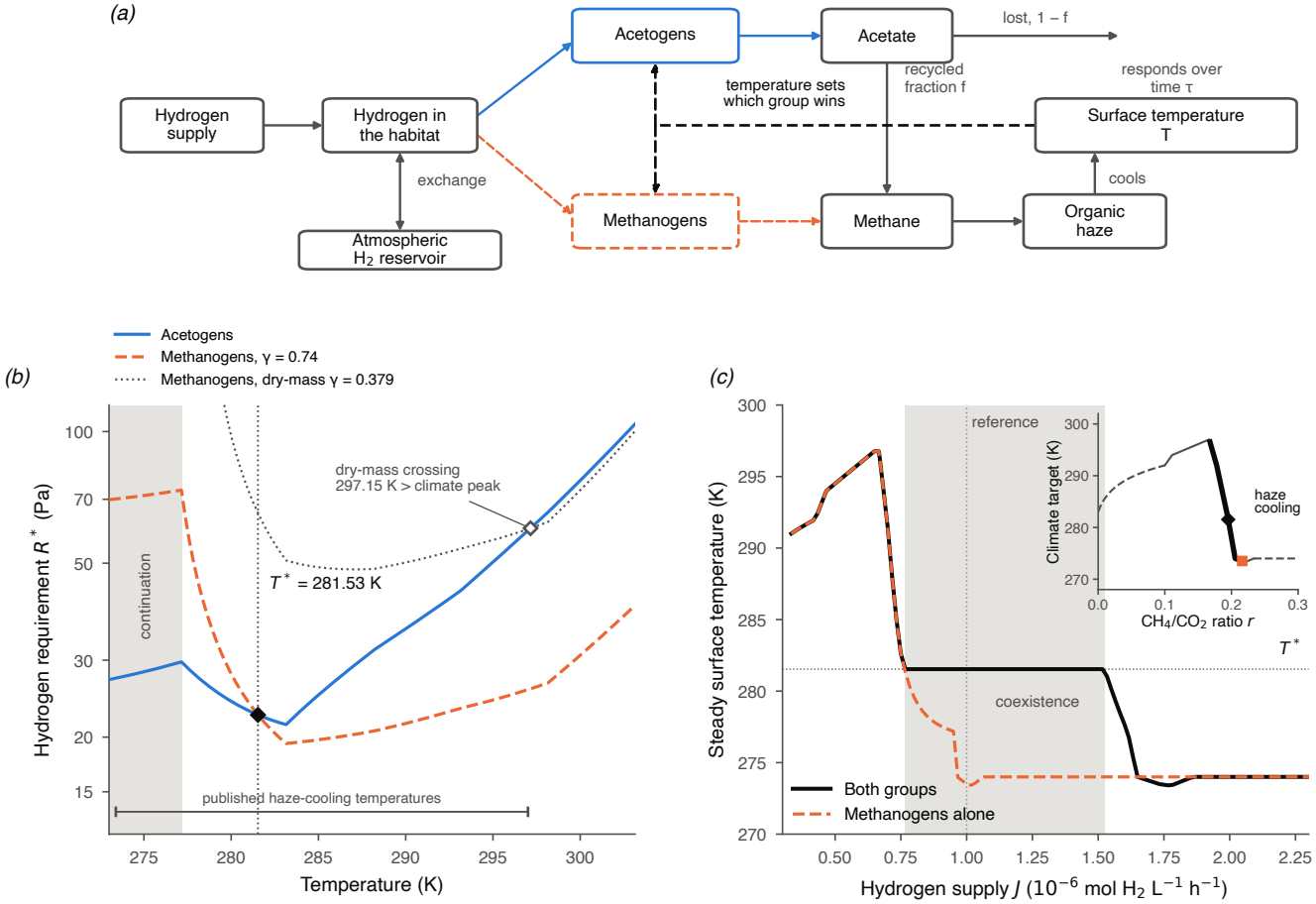
(a)
Acetogens
Acetate
lost, 1 – f
recycled fraction f
responds over time τ
temperature sets which group wins
Hydrogen supply
Hydrogen in the habitat
Surface temperature T
exchange
cools
Methanogens
Methane
Organic haze
Atmospheric H2 reservoir
(b)
Acetogens
Methanogens, γ = 0.74
Methanogens, dry-mass γ = 0.379
dry-mass crossing
297.15 K > climate peak
T* = 281.53 K
continuation
published haze-cooling temperatures
Hydrogen requirement R* (Pa)
Temperature (K)
(c)
reference
coexistence
haze cooling
Climate target (K)
CH4/CO2 ratio r
T*
Both groups
Methanogens alone
Steady surface temperature (K)
Hydrogen supply J (10−6 mol H2 L−1 h−1)